

Module 6: HMI Navigation & Fault Clearing

- PLC Tabletop Training Program · Lecture + Lab · 2.5 h
- v1.0 (April 2026)

Learning objectives

- Log into an HMI screen and navigate between key screens
- View and interpret active alarms and fault history
- Clear faults and reset the machine after a safe condition is restored
- Use the I/O status screen to verify field device communication
- Manually jog or operate outputs for maintenance testing

Overview

- Operating the HMI screen — login, alarm management, I/O status monitoring, and fault clearing procedures.

Lesson: HMI Overview & Login

- What an HMI is, screen navigation, and user access levels

What is an HMI?

- HMI stands for Human-Machine Interface. It's a touchscreen panel that lets the operator monitor and control the PLC system in real time. Common brands on site: Omron NB/NA series, Allen Bradley PanelView, AutomationDirect C-more.



Logging In

- Touch the screen to wake it — the home/overview screen will display
- Look for a padlock icon, user icon, or "Login" button (typically upper-right corner)
- Enter credentials using the on-screen keypad (check machine binder for default maintenance credentials)
- Press Enter or OK to log in

User Access Levels

- Table: Level | Typical Users | Permissions
- Operator | Production operators | View status, clear minor alarms, start/stop cycles
- Maintenance | Maintenance technicians | View I/O status, clear faults, manual jog, adjust setpoints
- Engineering | Controls engineers, automation lead | Full access — modify screens, change configurations, adjust safety parameters

Lesson: Alarm Management & Fault Clearing

- Reading alarms, understanding fault codes, and restoring normal operation

Viewing Active Alarms

- Navigate to the Alarms screen (often a bell icon or dedicated Alarms button). Active alarms display the alarm code, description, timestamp, and associated tag name. Most recent alarms appear at the top.

Fault Clearing Procedure

- Read the alarm — note the alarm code, description, and timestamp. Take a photo for documentation.
- Identify the root cause — use the alarm description and tag name to determine what triggered the fault. Open PLC software if needed to trace the associated logic.
- Correct the physical condition — clear an obstruction, replace a sensor, reset a safety device, fix a wiring issue. Never clear an alarm without addressing the cause.
- Clear the alarm on the HMI — select the alarm and press Clear/Reset. Some alarms require the fault condition to be gone before they can be cleared.
- Verify normal operation — confirm the alarm doesn't immediately return and the machine enters Ready state.
- [DANGER] Never 'Clear and Hope': Clearing an alarm without fixing the underlying cause is a recipe for recurring faults, production downtime, and potential safety incidents. If the alarm comes back after clearing, the root cause hasn't been resolved.

I/O Status Screen

- The I/O status screen shows real-time states of all inputs and outputs. Use it to verify sensors are communicating, outputs are energizing, and the physical world matches what the PLC reports.

Lesson: Manual Control & Setpoint Adjustment

- Jogging outputs, testing devices, and adjusting parameters from the HMI

Manual Jog/Output Control

- During maintenance, you may need to manually operate individual outputs — jog a motor, extend a cylinder, energize a solenoid — for testing or positioning. This is done from the HMI's Manual Control screen (maintenance access required).
- [WARNING] Safety First: Manual mode bypasses the normal automatic sequence. Ensure the area is clear, safety guards are functional, and all personnel are aware before jogging any output.

Setpoint Adjustment

- Timers, counters, speed references, and recipe parameters can be adjusted from the HMI. Always note the original value before changing it. Changes take effect immediately on most systems.

Lab: Live HMI Walkthrough

- After: HMI Overview & Login
- Duration: 10 min
- At the bench HMI, log in, navigate to key screens: overview, I/O status, alarms, manual control. Activate a proximity sensor and show the I/O status screen reflect it.
- Each student navigates to a different screen on the HMI. Students confirm they can find the I/O status screen and identify physical device states.

Lab: Fault Injection → Alarm → Clearance

- After: Alarm Management & Fault Clearing
- Duration: 15 min
- Trainer creates a fault (blocks area scanner). Walk through: read alarm → identify cause → clear obstruction → clear alarm → verify normal operation.
- After trainer demo, each student performs the full cycle: trainer injects a fault → student reads alarm → fixes physical cause → clears alarm → verifies.

Quiz

- What is the correct order for clearing a fault on the HMI?
- Clear the alarm first, then investigate the cause | Read the alarm → Identify and fix the root cause → Clear the alarm → Verify normal operation | Restart the PLC → Clear all alarms → Resume production | Call the supervisor and wait

Quiz

- The HMI shows three access levels. What can a Maintenance-level user do that an Operator cannot?
- View production statistics | Access I/O status, clear faults, manual jog, and adjust setpoints | Modify HMI screens | Nothing — they have the same access

Quiz

- An alarm appears on the HMI. You clear it, but it immediately returns. What does this mean?
- The HMI is malfunctioning | The root cause of the fault has not been resolved | You need to restart the PLC | The alarm has a 5-minute timeout